

Day 05 of 60

YES, YOU NEED MATH.

No, not all of it.

Here's the real list.

Stop wasting months on college math textbooks. You only need the intuition behind these specific concepts to build models.



3 Core Branches



What to learn



What to skip

Linear Algebra

Statistics

Calculus



Vinod Bavage
Machine Learning · AI

</> vinodbavage31

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Linear Algebra in 1 Slide

Everything in Machine Learning is a matrix. Datasets are matrices. Images are matrices. Neural Networks are just a series of matrix multiplications. You don't need to multiply by hand, but you **must** know how dimensions work.

What you actually need:

1 Vectors & Matrices

2 Dot Products

3 Matrix Inversion

4 Eigenvectors / Values



Statistics You Must Know

If you can't understand the data, you can't build a model. Statistics is the language of data. It tells you if your data is biased, how it's distributed, and whether your model's success was just luck.

What you actually need:

1

Mean & Variance

2

Distributions (Normal)

3

Bayes' Theorem

4

Hypothesis Testing



Calculus — Only This Bit

Models learn by minimizing error. Calculus helps us find the "bottom of the curve" (the lowest error). You don't need to solve complex integrals. You just need to understand **slopes**.

What you actually need:

1 Derivatives (Slopes)

2 Partial Derivatives

3 The Chain Rule

! Gradient Descent



Targeted Visual Resources

Don't read dense, dry textbooks. Visual intuition is 100x more valuable for applied Machine Learning.

Linear Algebra	3Blue1Brown YouTube: "Essence of Linear Algebra" Playlist
Statistics	StatQuest with Josh Starmer YouTube: "Machine Learning Fundamentals"
Calculus	3Blue1Brown YouTube: "Essence of Calculus" Playlist



THE MYTH

"You need a PhD in Mathematics to do Machine Learning."

THE TRUTH

You need high school math intuition, and you let **NumPy** and **PyTorch** do the actual calculations for you.

05

✓ Day 05 Complete

Tomorrow on Day 06:

Data Preprocessing Checklist

How to clean and prepare a dataset before feeding it to a model.

Follow for the full 60-day ML series

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